

Your grade: 80%

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1. Suppose your training examples are sentences (sequences of words). Which of the following refers to the j^{th} word in the i^{th} training example?

1 / 1 point

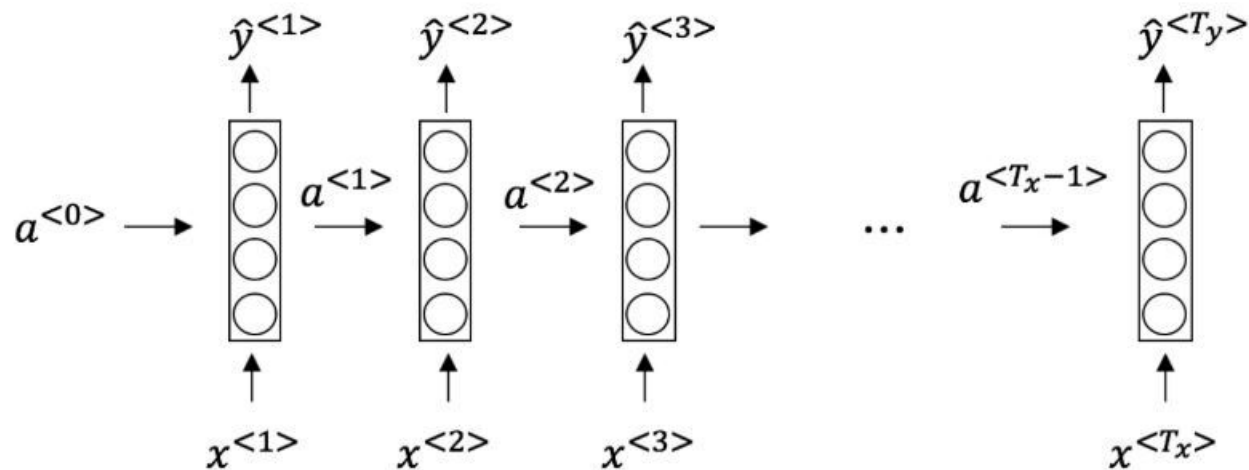
- ☒ $x^{(i)<j>}$
- ☐ $x^{<i>(j)}$
- ☐ $x^{(j)<i>}$
- ☐ $x^{<j>(i)}$

✓ **Correct**

We index into the i^{th} row first to get the i^{th} training example (represented by parentheses), then the j^{th} column to get the j^{th} word (represented by the brackets).

2. Consider this RNN:

1 / 1 point



This specific type of architecture is appropriate when:

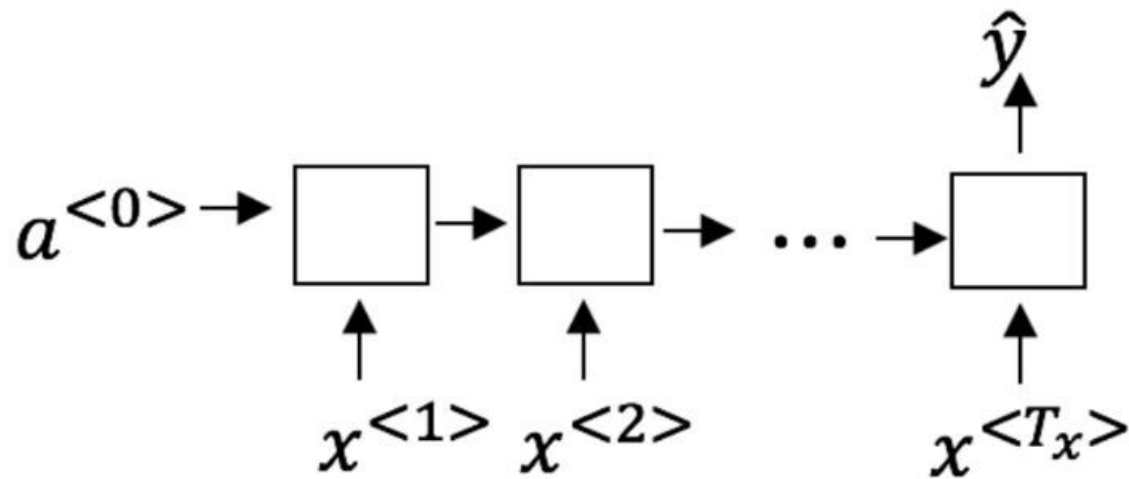
- ☒ $T_x = T_y$
- ☐ $T_x < T_y$
- ☐ $T_x > T_y$
- ☐ $T_x = 1$

✓ Correct

It is appropriate when every input should have an output.

3. To which of these tasks would you apply a many-to-one RNN architecture? (Check all that apply).

1 / 1 point



- ☐ Speech recognition (input an audio clip and output a transcript)
- ☒ Sentiment classification (input a piece of text and output a 0/1 to denote positive or negative sentiment)

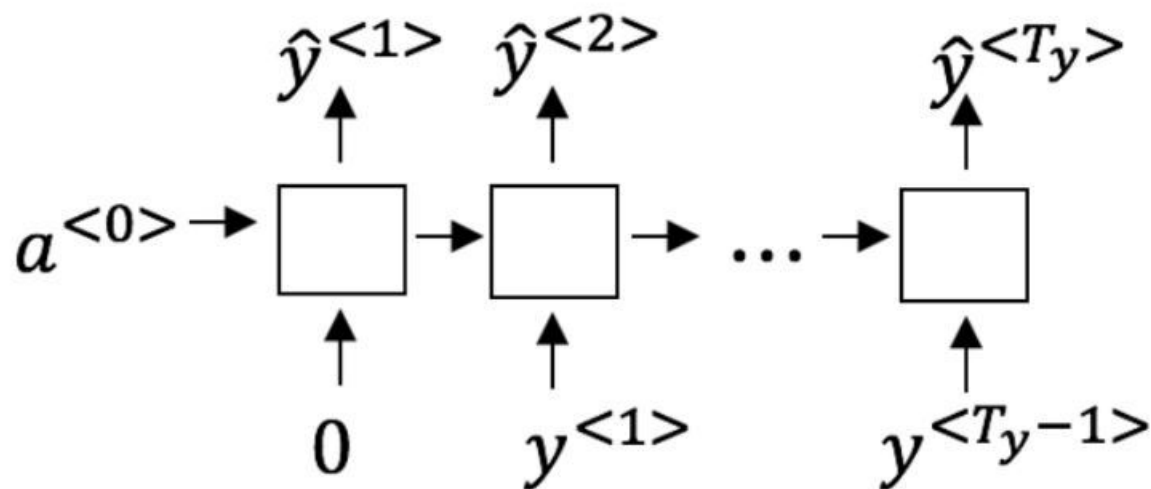
✔ **Correct**
Correct!

- ☐ Image classification (input an image and output a label)
- ☒ Gender recognition from speech (input an audio clip and output a label indicating the speaker's gender)

✔ **Correct**
Correct!

4. You are training this RNN language model.

1 / 1 point



At the t^{th} time step, what is the RNN doing?

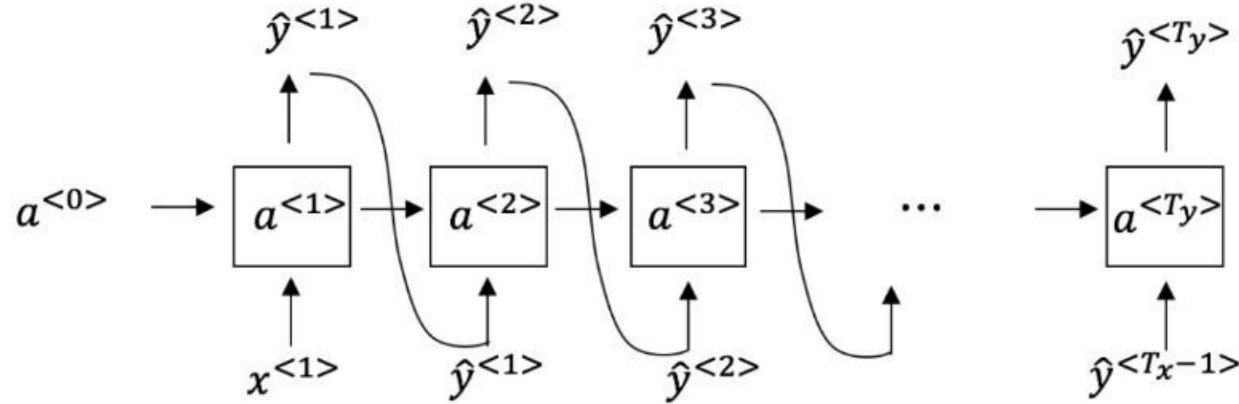
- ☐ Estimating $P(y^{<1>}, y^{<2>}, \dots, y^{<t-1>})$
- ☐ Estimating $P(y^{<t>})$
- ☒ Estimating $P(y^{<t>} \mid y^{<1>}, y^{<2>}, \dots, y^{<t-1>})$
- ☐ Estimating $P(y^{<t>} \mid y^{<1>}, y^{<2>}, \dots, y^{<t>})$

✓ **Correct**

Yes, in a language model we try to predict the next step based on the knowledge of all prior steps.

5. You have finished training a language model RNN and are using it to sample random sentences, as follows:

1 / 1 point



True/False: In this sample sentence, step t uses the probabilities output by the RNN to pick the highest probability word for that time-step. Then it passes the ground-truth word from the training set to the next time-step.

☒ False

☐ True

✓ **Correct**

The probabilities output by the RNN are not used to pick the highest probability word and the ground-truth word from the training set is not the input to the next time-step.

1 / 1 point

6. You are training an RNN model, and find that your weights and activations are all taking on the value of NaN ("Not a Number"). Which of these is the most likely cause of this problem?
- ☐ Vanishing gradient problem.
 - ☒ Exploding gradient problem.
 - ☐ The model used the ReLU activation function to compute $g(z)$, where z is too large.
 - ☐ The model used the Sigmoid activation function to compute $g(z)$, where z is too large.

✓ Correct

1 / 1 point

7. Suppose you are training an LSTM. You have a 10000 word vocabulary, and are using an LSTM with 100-dimensional activations $a^{<t>}$. What is the dimension of Γ_u at each time step?
- ☐ 1
 - ☒ 100
 - ☐ 300
 - ☐ 10000

✓ Correct

Correct, Γ_u is a vector of dimension equal to the number of hidden units in the LSTM.

1 point

8. True/False: In order to simplify the GRU without vanishing gradient problems even when training on very long sequences you should always remove the Γ_u . I.e., setting $\Gamma_u = 0$.

☒ False

☐ True

⊗ **Incorrect**

No, if $\Gamma_u \approx 0$ for a timestep, the gradient can propagate back through that timestep without much decay. For the signal to backpropagate without vanishing, we need $c^{<t>}$ to be highly dependent on $c^{<t-1>}$.

9. True/False: Using the equations for the GRU and LSTM below the Update Gate and Forget Gate in the LSTM play a role similar to $1 - \Gamma_u$ and Γ_u .

1 point

GRU

$$\tilde{c}^{<t>} = \tanh(W_c[\Gamma_r * c^{<t-1>}, x^{<t>}] + b_c)$$

$$\Gamma_u = \sigma(W_u[c^{<t-1>}, x^{<t>}] + b_u)$$

$$\Gamma_r = \sigma(W_r[c^{<t-1>}, x^{<t>}] + b_r)$$

$$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + (1 - \Gamma_u) * c^{<t-1>}$$

$$a^{<t>} = c^{<t>}$$

LSTM

$$\tilde{c}^{<t>} = \tanh(W_c[a^{<t-1>}, x^{<t>}] + b_c)$$

$$\Gamma_u = \sigma(W_u[a^{<t-1>}, x^{<t>}] + b_u)$$

$$\Gamma_f = \sigma(W_f[a^{<t-1>}, x^{<t>}] + b_f)$$

$$\Gamma_o = \sigma(W_o[a^{<t-1>}, x^{<t>}] + b_o)$$

$$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + \Gamma_f * c^{<t-1>}$$

$$a^{<t>} = \Gamma_o * \tanh c^{<t>}$$

☐ False

☒ True

✗ Incorrect

No. Instead of using Γ_u to compute $1 - \Gamma_u$, LSTM uses 2 gates (Γ_u and Γ_f) to compute the final value of the hidden state. So, Γ_f is used instead of $1 - \Gamma_u$.

1 / 1 point

10. Your mood is heavily dependent on the current and past few days' weather. You've collected data for the past 365 days on the weather, which you represent as a sequence as $x^{<1>}, \dots, x^{<365>}$. You've also collected data on your mood, which you represent as $y^{<1>}, \dots, y^{<365>}$. You'd like to build a model to map from $x \rightarrow y$. Should you use a Unidirectional RNN or Bidirectional RNN for this problem?

- ☐ Bidirectional RNN, because this allows the prediction of mood on day t to take into account more information.
- ☐ Bidirectional RNN, because this allows backpropagation to compute more accurate gradients.
- ☐ Unidirectional RNN, because the value of $y^{<t>}$ depends only on $x^{<t>}$, and not other days' weather.
- ☒ Unidirectional RNN, because the value of $y^{<t>}$ depends only on $x^{<1>}, \dots, x^{<t>}$, but not on $x^{<1>}, \dots, x^{<365>}$.

✓ Correct